Team Control Number
12461

Problem Chosen
B
2022
HiMCM/MidMCM
Summary Sheet

Analysis and study of atmospheric CO_2 concentration

Summary

With the continuous acceleration of industrialization, a large number of fossil fuels are burned, resulting in a trend of increasing carbon dioxide concentration in the atmosphere year by year. **Effectively analyzing and predicting** the concentration of carbon dioxide in the atmosphere and its impact on the earth's climate temperature is conducive to formulating corresponding strategic plans for the improvement of the future ecological environment. In this paper, the concentration of carbon dioxide in the atmosphere and the change of sea and land temperature over time are analyzed by using relevant **predictive models**, and the relationship between concentration and sea temperature is analyzed.

Part A of Problem 1 establishes a mathematical analytical calculation model based on **average growth rates**. the average growth rate of each period was calculated by dividing it into four periods of 10 years, it was concluded that **the carbon dioxide content in March 2004 did not increase significantly in the average rate of the decade**. Part B establishes **a time series, neuron network and function fitting** forecast model to analyze the relationship between the concentration of carbon dioxide in the atmosphere and time. In Part C, we bring the time point of 2100 into the three prediction models mentioned above, and obtain that the carbon dioxide concentration under the three models is **581.24ppm, 584.95ppm and 534.40ppm**, respectively. At the same time, through the calculation and analysis of the prediction model, the concentration of carbon dioxide in the atmosphere will **reach 685 ppm around 2135**. Part D begins by defining the evaluation criteria of 4 predictive models: **accuracy, precision, recall, and ROC curve**. After analyzing and calculating the above four evaluation criteria one by one, the comprehensive analysis shows that the prediction model under **the neuron network is the most accurate**, followed by the time series, and the poor accuracy is the function fitting.

Part A of problem 2 analyzes the relationship between time and mean land and ocean temperature based on the overall data by establishing **a quadratic polynomial regression model**. Then, the local values from 1959 to 1980 were selected as the reference period, the actual data and regression data were calculated and compared to test the accuracy of the model, and then **the fitting function** was used to predict the future average land and ocean temperature, and it was solved that it would change by **1.25°C in 2027, 1.50°C in 2036, and 2°C in 2050**. In Part B, we perform **linear fitting** by scatterplot of carbon dioxide concentration and sea and land temperature, and after fitting the data, calculate and **draw the residual map of the fitted model**, confirm the feasibility of the model, and conclude that the increase of land and ocean temperature leads to the increase of carbon dioxide concentration. Part C Based on the analysis of the mechanism of action of terrestrial and marine ecosystems, we conclude that the relationship between land-ocean temperature and carbon dioxide can be solved by the above model **in the short term of 20 years**.

Keywords：Average growth rate　Time series　Neuron networks　Polynomial regression analysis　ROC curve

Contents

1 Introduction 4
1.1 Problem Background 4
1.2 Restatement of the Problem 4
1.3 Our Work 4
2 Assumptions and Justifications 6
3 Notations 6
4 ModelI:Prediction model of carbon dioxide concentration 6
4.1 Model preprocessing 6
4.1.1 Determination of the calculation period 6
4.1.2 Definition of average growth rate 6
4.2 The Establishment of Model 1 7
4.2.1 Time Series Forecasting Models 7
4.2.2 Neural Network Models 8
4.2.3 Linear Fitting Model 9
4.3 Prediction Model Accuracy Calculation 9
4.3.1 Accuracy Calculation 10
4.3.2 Accuracy Calculation 10
4.3.3 Recall Calculation 10
4.3.4 ROC curve analysis 10
4.4 The Solution of Model 1 11
4.4.1 Question 1a Calculation result 11
4.4.2 Question 1b calculation result 11
4.4.3 Question 1c calculation results 12
4.4.4 Question 1d calculation result 12
5 ModelII: Model of temperature vs. carbon dioxide concentration ... 13
5.1 The Establishment of Model II 13
5.1.1 Establish the relationship between time and land-sea mean temperature 13
5.1.2 Establish the relationship between temperature and carbon dioxide concentration 14
5.1.3 Time as a function of land-sea mean temperature 14
5.1.4 Carbon dioxide concentration as a function of land-sea mean temperature . 15
5.1.5 Model reliability discussion 15
6 Model Evaluation and Further Discussion 16
6.1 Strengths 16

6.2 Weaknesses 16
7 Non-technical articles 17
8 References 18
Appendices 19

1 Introduction

1.1 Problem Background

With the continuous development of human science and technology and the substantial improvement of social productivity, from the late industrial revolution to the present, production plants have been built on a large scale around the world, resulting in the increasing use of fossil fuels and the increasing concentration of carbon dioxide in the atmosphere, as of May 2022, the global average monthly concentration of atmospheric carbon dioxide has reached 421ppm, and the global warming trend is becoming more and more clear, so it is of great practical significance to analyze and predict the impact of future changes in atmospheric carbon dioxide concentration on temperature and ecological environment.

1.2 Restatement of the Problem

Based on the requirements of problem one, the following four aspects need to be solved:

- Analyse changes in CO2 concentrations on a 10-year cycle basis to see whether the increase in CO2 concentrations over the decade as of March 2004 is greater than the increase in the previous decade.
- Establish a variety of mathematical models based on the data sets in the table to analyze the past and predict the future concentration level of atmospheric carbon dioxide.
- Using the relevant models established above, the atmospheric carbon dioxide concentration in 2100 is predicted, and the time for the atmospheric carbon dioxide concentration to reach 685 ppm in each forecast model is given.
- Conduct relevant analysis of the accuracy of each model established.

Combined with the first question model, according to dataset 2, the relationship between land-ocean temperature and carbon dioxide concentration is analyzed, and the following three aspects of problem 2 are solved:

- Based on known data, establish a suitable model to predict future land and ocean temperature changes, and give the predicted time for temperature changes to reach 1.25°C, 1.50°C and 2°C.
- Combined with the above two types of data sets, a correlation model was constructed to analyze and demonstrate whether there was a correlation between carbon dioxide concentration and land and ocean temperature.
- Analyze the reliability and predictive power of the established model theory on the relationship between land-sea temperature and carbon dioxide concentration.

1.3 Our Work

In response to question 1a to analyze whether the increase in CO2 concentration in March 2004 was higher than the increase in CO2 concentration observed over the previous 10 years, we first use the average increase in CO2 concentration to reflect the change in atmospheric CO2 concentration during each decade. Calculating and analysing whether the local point in time in March 2004 had a certain impact on the phase of the ten-year cycle in which it was located, and extending the impact of March 2004 on carbon dioxide concentrations to the full year of 2004 on the stage of the ten-year cycle in which it was located, by calculating and comparing the average growth rate of carbon dioxide for the decade period in 2004 and

excluding 2004, respectively, To determine whether the increases in CO2 concentrations in March 2004 were higher than the increases observed in the previous 10 years.

In response to question 1b, a variety of mathematical models are required to fit historical and predict the concentration of carbon dioxide in the atmosphere in future years based on the data given in the question. From the perspective of time, this paper first uses time series to predict and analyze the concentration of carbon dioxide in the atmosphere. At the same time, an adaptive neuronal network model is used to predict the change of carbon dioxide concentration. In order to make better use of the data set given in the question, this paper finally uses the least squares method for linear fitting analysis and prediction. Therefore, time series, neuron networks, and linear fitting are established below to serve as mathematical models to reflect past and future changes in atmospheric carbon dioxide concentrations.

In response to the sub-question of question 1c, the three prediction models established in Part B were used to predict the atmospheric carbon dioxide concentration in 2100, and the carbon dioxide concentration in 2050 under the three prediction models was calculated, and compared with the predicted carbon dioxide concentration value of 685PPM in 2050 given in the question. The following method of variable iteration is used to calculate the prediction results under the three models. The differences in the prediction results are analyzed one by one and the relevant influencing factors that may occur.

In response to the question 1d, which of the three mathematical prediction models established in this paper is the most accurate, by finding out the common characteristics of the above three prediction models. Four evaluation indicators are given below to judge the accuracy of the above three prediction models: accuracy, recall, PR curve, and ROC curve; Through the comprehensive evaluation of these four evaluation indicators, a model with more accurate predictions in the above three models is finally obtained.

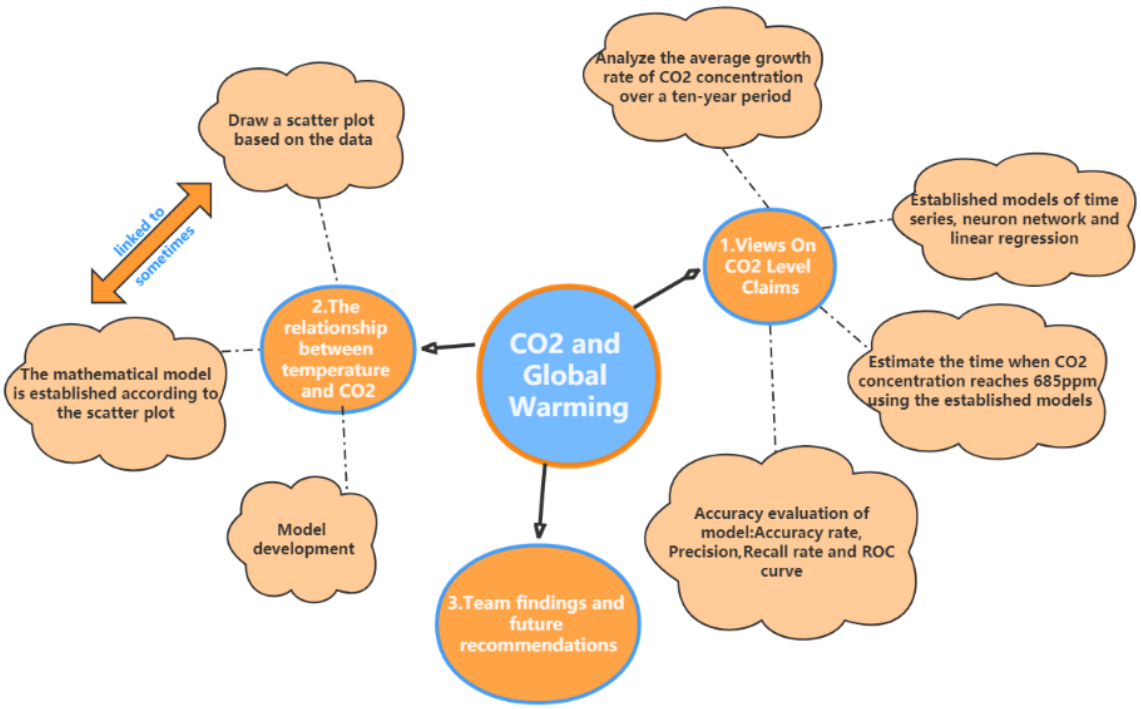


Figure 1:The flow chart of our work

Based on the second problem, we analyze the relationship between carbon dioxide concentration and land and ocean temperature based on the change trend of land and ocean temperature, draw a scatter plot based on the given data, and construct a mathematical model according to the trend of graph change. The specific idea is as shown above:

2 Assumptions and Justifications

1. Assume that the CO2 concentration is not affected by other interfering factors;
2. Assume that carbon dioxide concentrations and land-sea temperatures are affected only by the previous year;
3. Assume that changes in land-ocean temperatures are independent of Earth activity;
4. Only analyze the relationship between carbon dioxide and land-ocean temperatures.

3 Notations

The key mathematical symbols used in this paper are shown in Table 1.

Table 1: Notations used in this paper

Symbol	Definition	Unit
$\overline{\alpha}_{\mathrm{i}}$	Average growth rate of atmospheric CO2 concentration in the ith 10-year segment	\
C_{i}	Atmospheric CO2 concentration in year i (defined as 1964)	PPM
$C_{(\mathrm{t})}$	CO2 concentration as a function of time	\
δ	Residuals (predicted - actual)	\
β	Impact factor	\
$T_{(\mathrm{t})}$	Land and ocean temperature as a function of time	\
$C_{(T)}$	Carbon dioxide concentration as a function of time	\

4 Model Ⅰ:Prediction model of carbon dioxide concentration

4.1 Model preprocessing

4.1.1 Determination of the calculation period

According to the data given by data set 1, the historical data of CO2 concentration distribution in the atmosphere in 1964~2004 were selected for analysis. Firstly, the 1964~2004 year cycle is divided into 4 time periods, which are named I., II., Ⅲ, IV., and the 2004 data in IV. (1994-2004) are removed, and the time period from 1994 to 2003 is recorded as V.

4.1.2 Definition of average growth rate

Defining the concentration of CO2 in air in the ith year as , then (i+1) year is , then the average increase in CO2 concentration between the ith and i+1st years is:

$$\alpha_{\mathrm{i}} = \frac{C_{i+1} - C_i}{C_i} \times 100\% \tag{1}$$

The overall growth rate of atmospheric carbon dioxide concentration during the first time period (1964-1974). $\alpha_{\mathrm{I}} = \sum_{i=1}^{10} \alpha_i$, Then in time period I, the average growth rate :

$$\overline{\alpha}_{\mathrm{I}} = \frac{1}{10} \sum_{i=1}^{10} \alpha_i \tag{2}$$

At the same time, define the impact factor $\beta = \left| \frac{\overline{\alpha}_V}{\overline{\alpha}_{IV}} \right|$ （ β :Dimensionless ） . If $\beta \in [0.8,1.5]$, It can be approximated that the removal of CO2 concentration data for 2004 and the addition of CO2 concentration data for 2004 have negligible effect on the average growth rate of period IV and period V.

4.2 The Establishment of Model 1

4.2.1 Time Series Forecasting Models

For the sample data given by the title, set the year variable as t，The carbon dioxide concentration variable is C_{t} , Remember the two-dimensional totality（t, C_t）The sample observations are：

$$(\mathrm{t}_1, C_1), (t_2, C_2), \ldots\ldots (\mathrm{t}_n, C_n) \tag{3}$$

The univariate sample data of each component t and C can be obtained $\mathrm{t}_1, t_2 \ldots\ldots \mathrm{t}_n$ and C_1, $C_2, \ldots\ldots C_{\mathrm{n}}$ 。

Step 1: Stationary Daniel test

C_1, $C_2, \ldots\ldots C_{\mathrm{n}}$: The rank statistic for is $R_{(C_1)}$, $R_{(C_2)}, \ldots\ldots R_{(C_{\mathrm{n}})}$, The Spearman correlation coefficient is used to define the correlation coefficient for these two sets of rank statistics, the Spearman correlation coefficient：

$$\mathrm{q}_s = 1 - \frac{6}{n(n-1)} \sum_{i=1}^{n} d_i^2, d_i = t - R_t, i = 1,2\ldots\ldots, \ \mathrm{n} \tag{4}$$

Then, do the hypothesis test:

$$H_0\text{:} \ \rho_{XY} = 0 \qquad H_1\text{:} \ \rho_{XY} \neq 0 \tag{5}$$

When established, statistics $T = \frac{\mathrm{q}_{XY}\sqrt{n-2}}{\sqrt{1-q_{XY}^2}}$ T distribution t(n-2) with degrees of freedom (n-2).

For a given significance level α , while $|T| \leq \mathrm{t}_{\frac{\alpha}{2}}(n-2)$, that is, the sequence is stable.

Step 2: Covariance function and autocorrelation function estimation

For stationary sequences of carbon dioxide concentrations $C_{\mathrm{t}} : C_1$, $C_2, \ldots\ldots C_{\mathrm{n}}$, Use the sample mean to estimate the mean of the random sequence：

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} C_t = \overline{C} \tag{6}$$

Calculated for the estimate of γ_{t} :

$$\hat{\gamma}_k = \frac{1}{n} \sum_{t=1}^{n-k} (X_{t+k} - \overline{X})(X_t - \overline{X}), 0 \leq \mathrm{k} \leq K \tag{7}$$

For the estimate calculation for ρ_{t} :

$$\hat{\rho}_k = \frac{\hat{\gamma}_k}{\hat{\gamma}_0}, 0 \le k \le K \tag{8}$$

Step 3: Differentialize sequences

If the sequence is not stationary, a differential operation is required:

$$\nabla^d X_t = \sum_{i=0}^{d} (-1)^i C_d^i X_{t-i} \tag{9}$$

Get a new series, observe whether its scatter plot is stable, and if not, repeat this step.

Step 4: The parameters(p、q)are determined

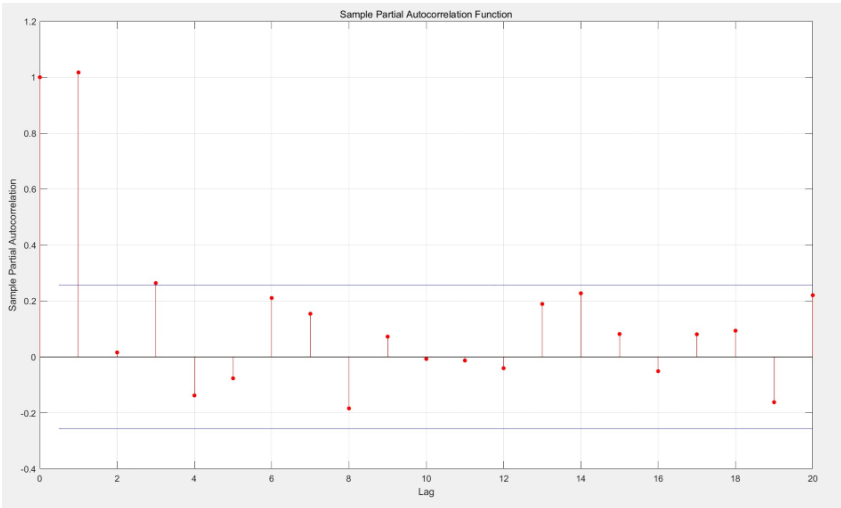


Figure 2 PACF of The Sample Data

Neural Network ModelsThrough the PACF plot of the given data sample, the time series is judged and the AR model is used.

4.2.2 Neural Network Models

Step 1:Establish BP neural network structure

BP neural network is a multi-level neural network with three or more layers, consisting of an input layer, an implied layer and an output layer. The BP neural network in this paper is trained in a gradient descent learning mode, that is, the historical data of atmospheric CO2 concentration from 1959 to 2001 is provided to the training network. According to the principle of reducing the error between the desired output and the actual output.

The following figure shows the BP neural network topology under this criterion:

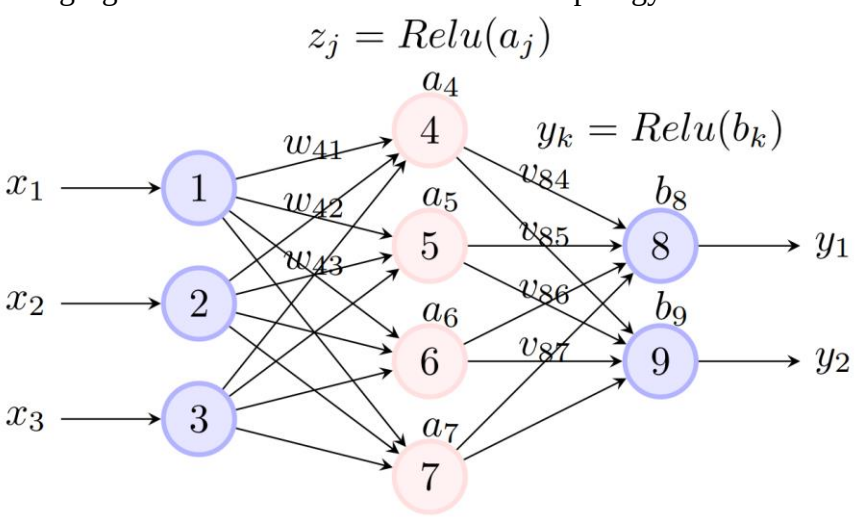


Figure 3 Topology of BP Neural Network

The number of layers of BP neural network is four layers, the number of neurons is 1,200,50,1, the activation function used is the Relu function, and the optimizer is selected as adam, where the parameters of adam are the learning rate of 0.001, betas=(0.9, 0.99), and the training times are 10000.

$$R=\begin{bmatrix}\varphi_1(\mathrm{t}_1) & \varphi_2(\mathrm{t}_1) & \cdots & \varphi_m(\mathrm{t}_1)\\ \varphi_1(\mathrm{t}_2) & \varphi_2(\mathrm{t}_2) & \cdots & \varphi_m(\mathrm{t}_2)\\ \vdots & \vdots & \ddots & \vdots\\ \varphi_1(\mathrm{t}_n) & \varphi_2(\mathrm{t}_n) & \cdots & \varphi_m(\mathrm{t}_n)\end{bmatrix},\quad A=\begin{bmatrix}a_1\\ a_2\\ \vdots\\ a_n\end{bmatrix},\quad C=\begin{bmatrix}C_1\\ C_2\\ \vdots\\ C_\mathrm{n}\end{bmatrix} \tag{14}$$

Step 2: Construct training samples

There are 63 years of historical data from 1959 to 2021, and this model first outputs all 63 sets of historical data. Adjust the number of nodes according to the actual results, observe the output results, and use this as the initial output; Then, the data in the sample is adjusted, appropriately culled, secondary input and secondary output are performed, and the output result in this case is compared with the initial output result to finally determine the training sample set.

4.2.3 Linear Fitting Model

Construct a linearly independent function sum $\{\varphi_\mathrm{k}(x)\,|\,k=1,2,\ldots\ldots,\ \mathrm{n}\}$。Under the question, a set of two-dimensional data $(\mathrm{t},C_{(t)})$, Seek a function $C(\mathrm{t})$, Make $C(\mathrm{t})$ closest to all data points under a certain criterion, that is, the function relationship fits well.

$\delta_\mathrm{i}=C(t)=C_i(i=1,2,\ldots\ldots,\ \mathrm{n})$, Call δ_i the deviation of the fitting function $C(\mathrm{t})$ at point t, and to meet the requirements, we should make: $J=\sum_{i=1}^{n}(C_{(\mathrm{t}_i)}-C_i)^2$ The minimum value is reached.

$$\mathrm{f}(k)=\sum_{i=1}^{m}a_k\varphi_k(x) \tag{10}$$

Substituting the above equation, then $J=J\ (a_1,a_2,\ldots\ldots,\ a_n)$ It's about $a_1,a_2,\ldots\ldots,\ a_m$ The multivariate function consists of:

$$\frac{\alpha_J}{\alpha_{a_k}}=0,k=1,2,\ldots\ldots,\ \mathrm{m} \tag{11}$$

$$\sum_{i=1}^{n}((C_{(\mathrm{t}_i)}-C_i)\bullet\varphi_k(t_i))=0,\ k=1,2,\ldots\ldots,\ \mathrm{m} \tag{12}$$

$$\sum_{j=1}^{n}(\sum_{i=1}^{n}\varphi_\mathrm{j}(t_i)\cdot\varphi_k(t_i))\cdot a_j=\sum_{i=1}^{n}C_i\varphi_k(t_i),k=1,2,\ldots\ldots,\ \mathrm{m} \tag{13}$$

Note that the system of normal equations is:

$$R^TRA=R^TC \tag{15}$$

When R is to full rank , R^TR is invertible and gives the system of equations a unique solution.

4.3 Prediction Model Accuracy Calculation

First, the above three prediction models can depict the concentration of carbon dioxide in the atmosphere from 1959 to 2021.

The defined residuals are:

$$\delta = \text{Predicted - Actual value} \tag{16}$$

Define a series of residual ranges as follows, followed by time series, neuron networks, and univariate function fitting:

$$\begin{aligned}
&\text{TP（Genuine）：} \delta \in [0,0.68],\ \delta \in [0,0.52],\ \delta \in [0,0.54] \\
&\text{FP（False positive）：} \delta \in (0.68,1.35],\ \delta \in (0.52,1.02],\ \delta \in (0.54,0.65] \\
&\text{FN（False negative）：} \delta \in [-0.75,-0.30),\ \delta \in [-0.55,-0.32),\ \delta \in [-3.51,-1.48) \\
&\text{TN（Really negative）：} \delta \in [-0.30,0),\ \delta \in [-0.32,0),\ \delta \in [-1.48,0)
\end{aligned} \tag{17}$$

4.3.1 Accuracy Calculation

Accuracy refers to the number of samples in the classification model that are predicted to be accurate/the total number of samples worth of observations for a given test data set. In this paper, we refer to the use of the above three prediction models for prediction, and the proportion of the number of samples with residuals within a certain range of predictions to the total data set, namely:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \times 100\% \tag{18}$$

4.3.2 Accuracy Calculation

Accuracy measures the accuracy of the detection system, the number of samples correctly predicted by the model as positive / the number of samples predicted by the model as positive. In this article, the number of samples with residuals greater than zero and closer to the actual value is the proportion of all samples with residuals greater than zero, that is:

$$PRECISION = \frac{TP}{TP + FP} \times 100\% \tag{19}$$

4.3.3 Recall Calculation

Recall is also sensitivity, which measures the recall rate of the detection system, the number of positive samples that the model predicts correctly/the number of positive samples in the actual observed data. In this context, the number of samples with positive residuals and predicted values closer to the actual value is the proportion of the total number of samples with all predicted values closer to the actual value, that is:

$$RECALL = \frac{TP}{TP + FN} \times 100\% \tag{20}$$

4.3.4 ROC curve analysis

The definition of the true rate is:

$$TRUE\ \ POSITIVE\ \ RATE = \frac{TP}{TP + FN} \tag{21}$$

The false positive rates are:

$$FALSE\ \ POSITIVE\ \ RATE = \frac{FP}{TN + FP} \tag{22}$$

When measuring the generalization performance and accuracy of a model, the samples are sorted according to the prediction results of the model, and the samples are output as positive examples one by one in this order, and the true rate TPR and false positive rate FPR of the test sample are calculated each time, and these two are used as the vertical and horizontal

axes of ROC, so as to make an ROC curve. When the ROC curve in each prediction model is close to point (1,0), that is, the closer to the upper left for the ROC curve, the better the model performance, and also indicates the better the model generalization performance and the higher the accuracy. In addition, if the ROC curve of one model is completely wrapped by the curve of another model, it can be judged that the performance of the latter is better than the former.

4.4 The Solution of Model 1

4.4.1 Question 1a Calculation result

Table 2: Average Growth Rate

Period	Ⅰ	Ⅱ	Ⅲ	Ⅳ	Ⅴ
Aveage growth rate of CO_2 concentration	0.33%	0.45%	0.45%	0.56%	0.57%

From the above table, it can be concluded that the average growth rate of atmospheric carbon dioxide in each decade shows a gradual increase trend, which also indicates that the concentration of carbon dioxide in the atmosphere has been increasing since the 60s of the last century. At the same time, for the average growth rate of the fourth period from 1994 to 2004, it is calculated that when the 2004 carbon dioxide concentration data is used, the average growth rate of the carbon dioxide concentration in the fourth period is 0.56%, and when the carbon dioxide concentration data for 2004 is excluded, the average growth rate of the carbon dioxide concentration in the fifth period is 0.57%; This set of data illustrates the negligible effect of atmospheric carbon dioxide concentrations in 2004 on the average growth rate for the fourth time period in which it resides. Thus, it is indirectly stated that the March 2004 CO2 emissions did not fundamentally lead to a greater increase than in the previous 10-year period, and the conclusion is not agreed that the increase in CO2 concentration in March 2004 was higher than the increase in CO2 concentration observed in the previous 10-year period.

4.4.2 Question 1b calculation result

By solving the above 3 mathematical prediction models: time series, neuron networks and function fitting models. To test whether these three predictive models can accurately describe the concentration of carbon dioxide in the atmosphere in a past year and the concentration level of carbon dioxide in the atmosphere in the future. We select the data from 2001-2011 of dataset 1 for verification, and calculate the absolute error of the results obtained under the three models, as shown in the following table:

Table 3: Absolute Errors Under The Three Models

Year	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011
Absolute error of the time series	0.72	0.64	0.60	0.30	0.09	0.19	0.14	0.00	0.69	0.42
Absolute error of the network of neurons	0.01	0.53	0.12	0.21	0.20	0.14	0.04	0.08	0.35	0.04
Univariate linear absolute error	3.26	2.34	2.24	1.57	1.07	0.76	0.56	0.37	0.48	0.62

As can be seen from the above table, the prediction results of the three prediction models are within a certain time range, and the model has a good prediction fitting effect.

4.4.3 Question 1c calculation results

According to the above three predicted models, in 2100, the concentration of carbon dioxide in the atmosphere was 581.24ppm, 584.95ppm and 534.40ppm under the time series, neuron network and function fitting model, respectively. This result shows that the results under the three prediction models are relatively close, and it is approximate that the concentration of carbon dioxide in the atmosphere in 2100 will be around 560 ppm. In addition, for whether the concentration of carbon dioxide in the atmosphere reaches 685ppm in 2050, this paper takes the time point of 2050 as the input of the above three prediction models, and obtains that the forecast result under the time series is 469.35ppm, the forecast result under the neuron network is 478.42ppm, and the prediction result under function fitting is 453.70ppm.

The results suggest that even by 2050, the concentration of carbon dioxide in the air will not reach 685 ppm under any of the above prediction models. As for when the concentration of carbon dioxide in the atmosphere can reach 685 ppm, we use the carbon dioxide concentration as the input of the model and the time point as the output. The final time series, neuron network, and function fitting output results are: 2139, 2128, 4052. It is obvious that the prediction results under the function fit are very different from the prediction results under the other two models.

4.4.4 Question 1d calculation result

According to the four model evaluation criteria in the above model establishment, the accuracy, precision, and recall of the four prediction models are as follows:

Table 4: Evaluation of The Accuracy of Four Types of Models

	Time series	Networks of neurons	Univariate linearity
ACC	60%	90%	50%
PRECISION	67%	87.5%	505
RECALL	67%	100%	20%

It can be seen from the four evaluation criteria of time series, neuron network, and univariate linear function fitting accuracy, precision, recall and ROC curve: the ROC curve of the neuron network is always above the ROC curve of the other two prediction models, and its accuracy and recall rate are also high, which fully illustrates that the forecast model under the neuron network is more accurate, followed by the time series, and the accuracy is poor is the function fitting, and the obtained ROC curve is shown on the right:

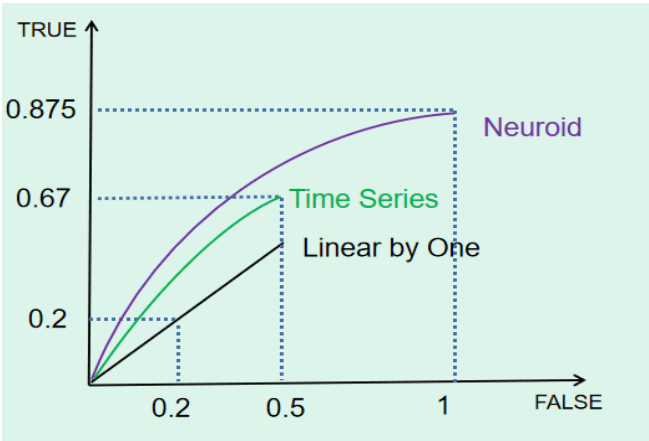


Figure 4 ROC Curve

5 Model Ⅱ: Model of temperature vs. carbon dioxide concentration

5.1 The Establishment of Model Ⅱ

5.1.1 Establish the relationship between time and land-sea mean temperature

Based on the analysis of the lower scatterplot and the associated rate of change, the quadratic polynomial of land and ocean temperature and time is $T = at^2 + bt + c$, where T is the land ocean temperature and T is the time scale.

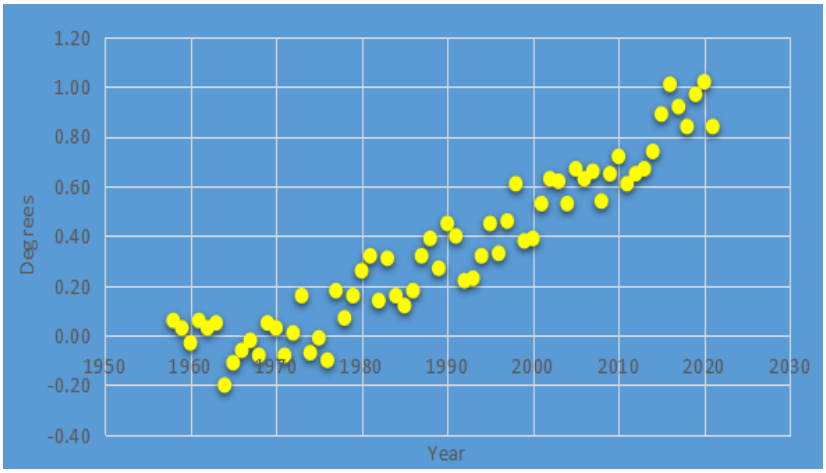


Figure 5 Scatter Plot of Land And Ocean Temperature From 1958 To 2021

The quadratic polynomial will be fitted using least squares, and the error function of the quadratic polynomial will be fitted Represented in matrix form:

$$\|t\omega - T\|_2 \tag{23}$$

Among them, the bias is fused into the parameters, $\omega = [c, b, a]$ Represents the matrix of coefficients for this quadratic polynomial, and the input matrix is as follows:

$$\begin{bmatrix} 1 & t_1 & t_1^2 \\ 1 & t_2 & t_2^2 \\ \vdots & \vdots & \vdots \\ 1 & t_n & t_n^2 \end{bmatrix} \qquad T = [T_1 \quad T_2 \quad \dots \quad T_n]^T \tag{24}$$

Therefore, the fitting problem is transformed into an optimization problem:

$$\min_{\omega} \|t\omega - T\|_2 \tag{25}$$

From the principle of extreme value solving, the matrix needs to be derived, namely:

$$\frac{\partial \|t\omega - T\|_2}{\partial \omega} = \frac{\partial \left(\omega^T T^T T\omega - T^T t\omega - \omega^T t^T T + T^T T\right)}{\partial \omega} \tag{26}$$
$$2t^T t\omega - 2t^T T = 0$$

This leads to:

$$\omega = \left(t^T t\right)^{-1} t^T T \tag{27}$$

That is, after finding the coefficient matrix of the quadratic polynomial between time and land and ocean temperature, the fitted quadratic function is obtained.

5.1.2 Establish the relationship between temperature and carbon dioxide concentration

Separate the historical data of carbon dioxide concentration and sea and land temperature from the data table given by the title and plot it in a scatterplot as a two-dimensional sample data set, as shown in the following figure:

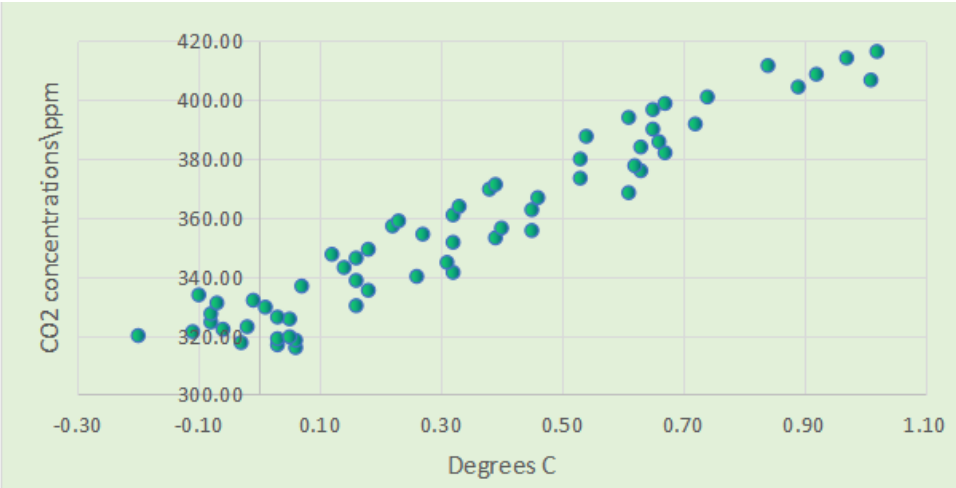


Figure5 Plot of Trends in CO_2 Concentrations And Land And Ocean Temperatures

It can be seen from the figure above that from 1959 to 2021, the carbon dioxide concentration increased linearly in terms of sea and land temperature, and since the linear fitting model was also introduced when analyzing the relationship between carbon dioxide concentration and time in part A of problem 1, the mathematical model for establishing a linear fit of a univariate function is not repeated in this part.The Solution of Model Ⅱ.

5.1.3 Time as a function of land-sea mean temperature

Based on the above model, the quadratic function relationship between time and land-ocean average temperature can be obtained as follows:

$$T(t) = 0.0001594t^2 - 0.6175t + 598 \tag{28}$$

According to the model and compared with the reference period from 1959 to 1980, the future average land and ocean temperature is projected, and it can be found that the average land and sea temperature will change by 1.25°C in 2027, 1.50°C in 2036, and 2°C in 2050. It can be seen that with the migration of time, the average land and ocean temperature is increasing faster and faster, and the combined effect of human activities and ecosystems has caused the intensification of global warming.

Table 5: Predicted And Actual Land And Ocean Temperatures

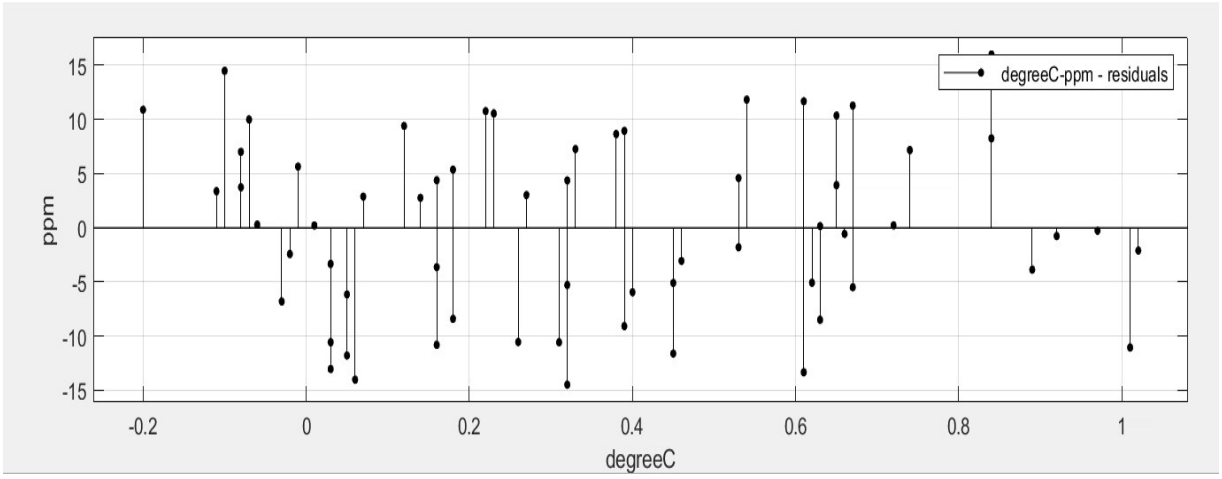
Year	Actual value	Predicted value
1959	0.04	0.03
1960	-0.05	-0.03
1961	0.06	0.06
1962	0.07	0.03
1963	0.07	0.05
1964	-0.08	-0.2
1965	-0.09	-0.11

1966	-0.10	-0.06
1967	0.11	-0.02
1968	-0.12	-0.08
1969	0.13	0.05
1970	0.14	0.03
1971	-0.15	-0.08
1972	0.16	0.01
1973	0.17	0.16
1974	0.19	-0.07
1975	0.20	-0.01
1976	-0.21	-0.1
1977	0.22	0.18
1978	0.23	0.07
1979	0.25	0.16
1980	0.26	0.26

5.1.4 Carbon dioxide concentration as a function of land-sea mean temperature

Through computational analysis, we find that the land-ocean average temperature does not increase with the increase of carbon dioxide concentration, but the carbon dioxide concentration increases with the increase of average land and ocean temperature, so the functional relationship between average land and ocean temperature and carbon dioxide concentration is solved as:

$$C(T) = 88.21T + 326.4 \tag{29}$$



At the same time, the residuals of the function model are calculated, and the residual plot is drawn as shown in the figure above, and the visible errors are within a controllable range, indicating that the model is established more accurately.

5.1.5 Model reliability discussion

According to the model built and the obtained function, and calculated and verified, in the short term, we predict that the model can predict the relationship between the two more accurately in the short term of about 20 years, that is, the model is more reliable in this time period, but in the long run, the model we build is not comprehensive enough and too idealistic to accurately predict subsequent data in the presence of multiple influencing factors. There

are many factors affecting the concentration of carbon dioxide in the atmosphere, and it can be seen that for ecosystems, the carbon sink function of terrestrial ecosystems is reduced, because warming reduces the net primary productivity of terrestrial ecosystems, increases iso-oxygen respiration and organic carbon decomposition; The decrease in the ocean's carbon dioxide absorption capacity may be related to the solubility of carbon dioxide, the decrease in seawater salinity, and the weakening or destruction of the ocean thermohaline circulation due to warming. The combination of these objective factors will cause the relationship between average land and ocean temperature and carbon dioxide to change from a certain period of time, so to build a long-term prediction model, we need to deeply consider all factors and improve them in real time, which is the solution and our problem.

6 Model Evaluation and Further Discussion

6.1 Strengths

- For the evaluation of model accuracy, we introduce multiple criteria of accuracy, accuracy, recall and ROC curve for multi-dimensional analysis, which makes our evaluation more scientific and accurate.
- Whether CO2 emissions in March 2004 led to a larger increase observed during the previous decade, we have segmented the data over a ten-year cycle to make the analysis more convincing and concise.
- This paper analyzes the trend of CO2 concentration changes in each decade by customizing the average CO2 growth rate over a ten-year period. At the same time, this paper also predicts the concentration of carbon dioxide under different time conditions by using time series, neuron networks, and function fitting. In addition, the relationship between sea and land temperature and time is fitted by scatter plot, and it is also concluded that carbon dioxide concentration changes with sea and land temperature. It is further explained that global warming has caused the concentration of carbon dioxide in the atmosphere to rise.

6.2 Weaknesses

- Short availability time period of the model; The effect of other substances in the atmosphere on carbon dioxide concentrations is not considered.
- The accuracy judgment of the model can be analyzed from more angles.

7 Non-technical articles

Low-carbon life--- protect the beautiful home

With the development of the world's industrial economy, the sharp increase in population, the infinite rise of human desire and the unrestrained production and lifestyle, the world climate is facing more and more serious problems, carbon dioxide emissions are getting larger and larger, the earth's ozone layer is suffering from an unprecedented crisis, global catastrophic climate change has repeatedly appeared, has seriously endangered the human living environment and health security, we can find that in recent years, large and small natural disasters have occurred from time to time around the world. Part of the reason for this may also be caused by the excessive emission of carbon dioxide by humans, so we must wake up and put low-carbon concepts into action.

The emergence of low-carbon living not only tells us what you can do to reduce carbon, but also tells people what they can do. Low carbon is a kind of life, habit, is a natural habit to save all kinds of resources around you, as long as you are willing to take the initiative to restrain yourself, improve your living habits, can also do a good job, as long as you can do more saving from the little things of life, no waste, will also make life better, lower carbon.

As long as we unite every force to reduce every gram of carbon emissions, ordinary people also have the power to change the future in the process of reducing carbon dioxide emissions. What are you still hesitating about, hurry up and join our actions, let's become low-carbon experts together!

8 References

[1] Guo Jijun, Guo Zhilong & Wang Ninglian. (2020). Study of Sunspot Activity Period in Earth Land-Ocean Temperature Index since 1880. Plateau Meteorological System (04), 851-858.

[2] Fan Yuexin, Xu Linglin, Liu Xiuming, Chen Guangshui, Yang Zhijie, Wang Jian. The relationship between atmospheric CO_2 concentration and global warming was analyzed from the perspective of terrestrial and Marine ecosystems [J]. Journal of Subtropical Resources and the Environment, 2012,7 (01): 55-60.

[3]He Jianghao, CAI Yulin, Qin Peng. Analysis of spatial and influencing factors of carbon dioxide [J]. Scientific notification,2020,65(Z1):194-202.

[4]Draper, N.R. and Smith, H. Applied Regression Analysis. Wiley Series in Probability and Statistics. 1998.

Appendices

Appendix 1

Partial code presentation

```
x1=xlsread('Data7.xlsx','B7:B47');
alpha1=[];
for i=1:40
    alpha=(abs(x1(i+1)-x1(i)))/x1(i);
    alpha1=[alpha1,alpha];
end
alpha11=(sum(alpha1(1:10)))/10;
alpha12=(sum(alpha1(10:20)))/10;
alpha13=(sum(alpha1(20:30)))/10;
alpha14=(sum(alpha1(30:40)))/10;
alpha2=(sum(alpha1(30:39)))/9;

CO2a=xlsread('Data7.xlsx','B4:B64')';
figure(1),parcorr(CO2a);%Draw the PACF plots for each model
Rt1=tiedrank(CO2a);%Seeking the rank of the original time series
nt1=length(CO2a);
t1=1:nt1;
Qs1=1-6/(nt1*(nt1^2-1))*sum((t1-Rt1).^2);%The Spearman's correlation coefficient was calculated
T1=Qs1*sqrt(nt1-2)/sqrt(1-Qs1^2);%The T statistic was calculated
t_1=tinv(0.975,nt1-2);%Upper alpha / 2 quantiles were calculated
CO2b=diff(CO2a);
m1=ar(CO2b,2,'ls');%The parameters of the model were estimated using the least squares method
fc1=predict(m1,CO2b');
fc1(end+1)=forecast(m1,CO2b',1);%A predictive value was calculated
fc2=[CO2a(1),CO2a+fc1'];%Seek the predicted value of the raw data
re1=abs((fc2(1:end-1)-CO2a)./CO2a);
residual1=CO2a-fc2(1:end-1);

%The functions were fitted and the prediction tests were performed
year=xlsread('Data7.xlsx','A2:A64');
year1=[ones(63,1) year];
ppm=xlsread('Data7.xlsx','B2:B64');
[b1,bint1,r1,rint1,stats1]=regress(ppm,year1);
for i=1:63
    ppm1(i)=b1(1)+b1(2)*year(i);
end
residual2=ppm'-ppm1;

residual11=abs(residual1(end-19:end));
residual21=abs(residual2(end-19:end));
residual12=residual1(end-19:end);
residual22=residual2(end-19:end);
error1=abs(error(end-19:end));
error2=error(end-19:end);

degreeC=xlsread('Data8.xlsx','B3:B65');
```

```
[b2,bint2,r2,rint2,stats2]=regress(ppm,degreeC);
ppm2=[];
for i=1:63
    ppm2(i)=326.4+88.21*degreeC(i);
end
residual3=ppm'-ppm2;
for i=1:63
    degreeC1(i)=598-0.6175*year(i)+0.0001594*year(i)*year(i);
end
residual4=degreeC'-degreeC1;
for i=1:22
    degreeC2(i)=598-0.6175*year(i)+0.0001594*year(i)*year(i);
end
degreeC3=degreeC(1:22)';
year2=year(1:22);
degreeC4=degreeC3';

%Neural network
% Solve an Input-Output Fitting problem with a Neural Network
% Script generated by Neural Fitting app
% Created 11-Nov-2022 13:12:24
%
% This script assumes these variables are defined:
%
%   year - input data.
%   ppm - target data.

x = year';
t = ppm';

% Choose a Training Function
% For a list of all training functions type: help nntrain
% 'trainlm' is usually fastest.
% 'trainbr' takes longer but may be better for challenging problems.
% 'trainscg' uses less memory. Suitable in low memory situations.
trainFcn = 'trainlm';  % Levenberg-Marquardt backpropagation.

% Create a Fitting Network
hiddenLayerSize = 10;
net = fitnet(hiddenLayerSize,trainFcn);

% Setup Division of Data for Training, Validation, Testing
net.divideParam.trainRatio = 70/100;
net.divideParam.valRatio = 15/100;
net.divideParam.testRatio = 15/100;

% Train the Network
[net,tr] = train(net,x,t);

% Test the Network
y = net(x);
e = gsubtract(t,y);
```

```
performance = perform(net,t,y)

% View the Network
view(net)

% Plots
% Uncomment these lines to enable various plots.
%figure, plotperform(tr)
%figure, plottrainstate(tr)
%figure, ploterrhist(e)
%figure, plotregression(t,y)
%figure, plotfit(net,x,t)
```